Chapter 2
IoT Protocols

IoT Protocols

- IoT networks have unique traffic characteristics, which are significantly different from traditional networks
- IoT devices are often constrained, have limited computing power, are powered by the battery, and communicate over lossy networks.
- Due to these reasons, the traditional TCP/IP protocol suite used in legacy networks has not automatically become the fundamental standard in IoT

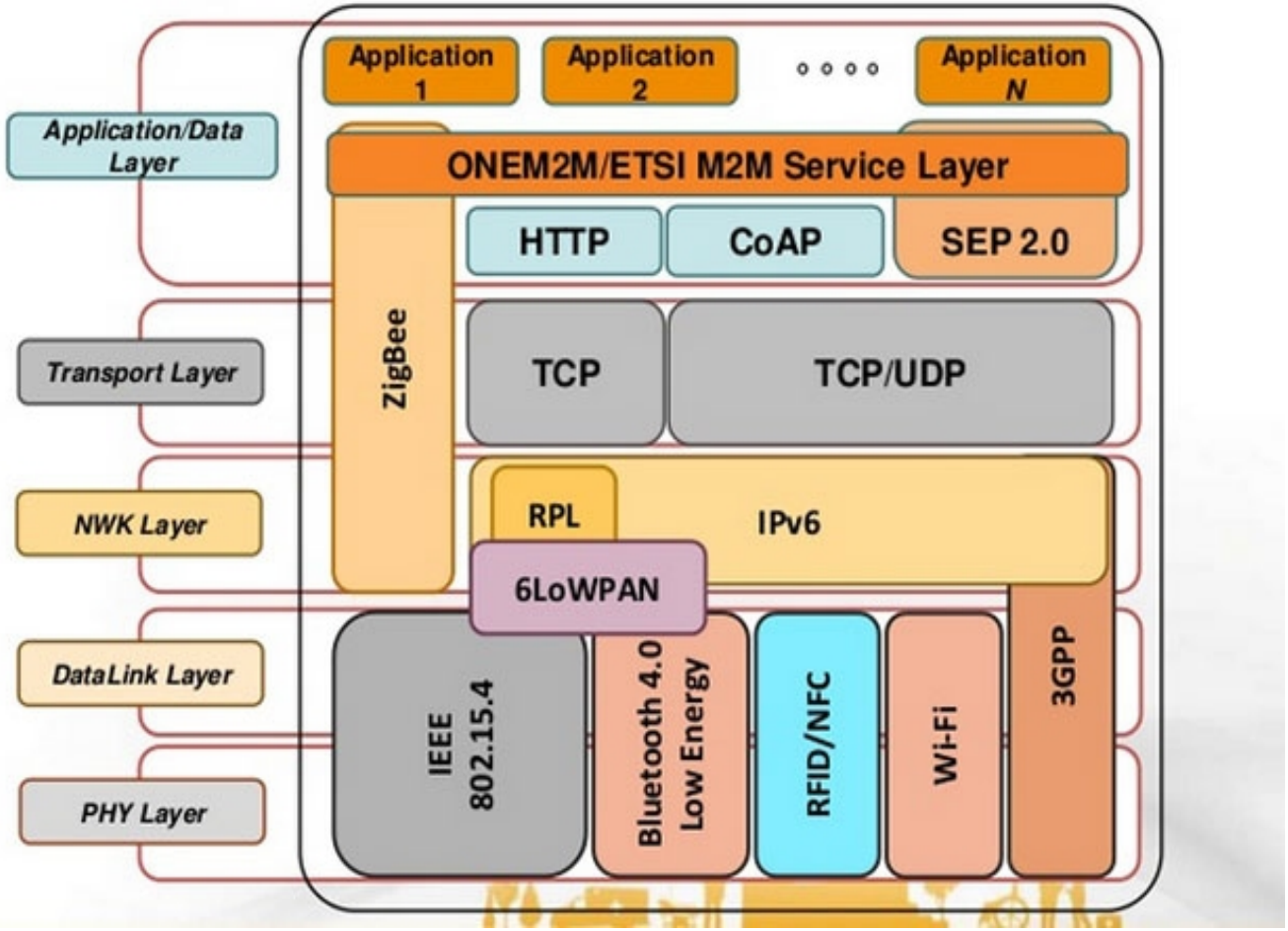
IoT Protocols

- The heterogeneity of IoT application domains also means that different use cases have different requirements, which can only hardly be addressed with one standard.
- The most popular way of connecting them is using wireless technologies.

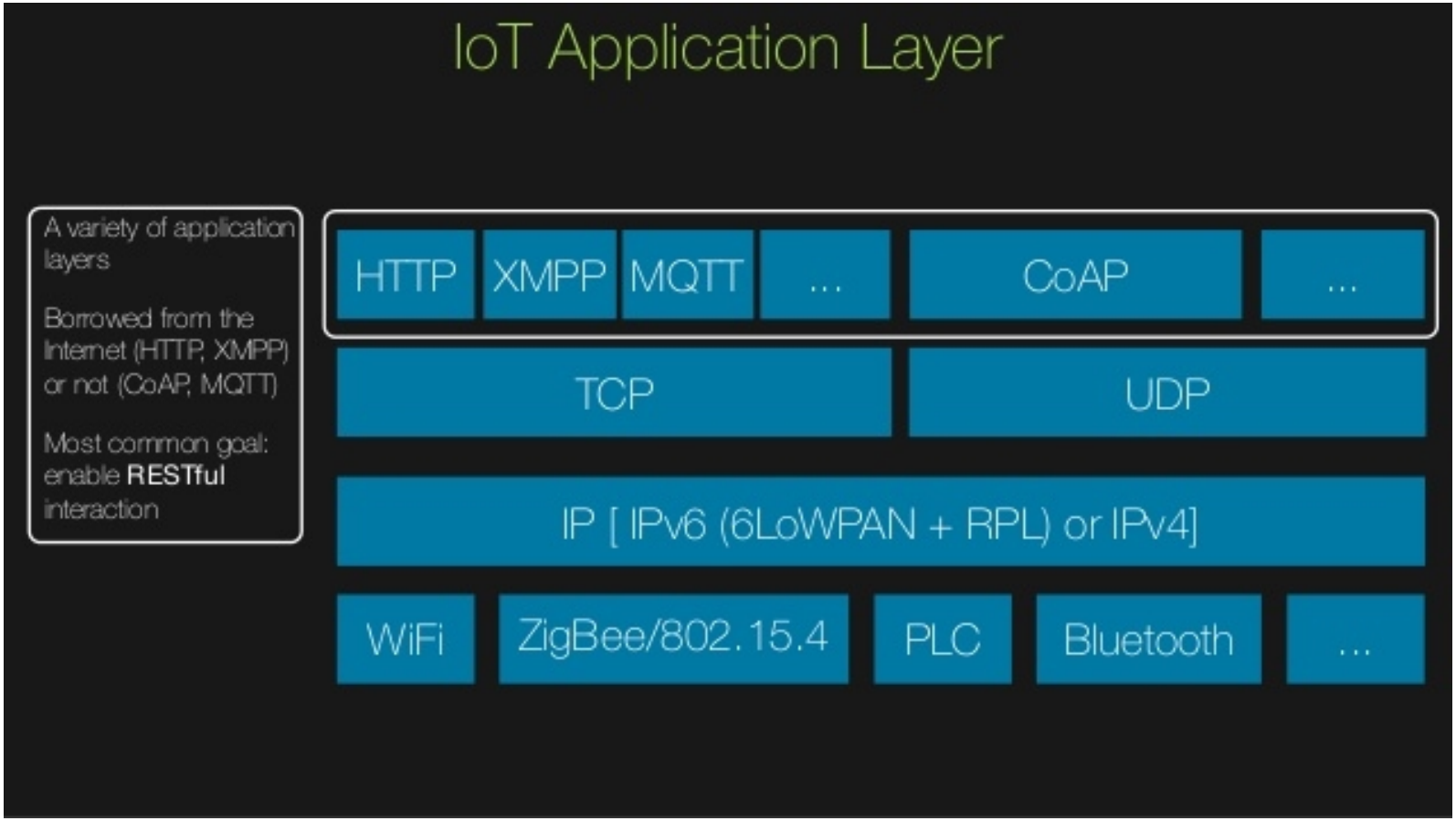
IoT Protocols

- One of the main IoT building blocks is a Wireless Sensor Network (WSN) or a Wireless Sensor and Actuator Network (WSAN)
- WSN needs to work with traditional network

IoT Protocols



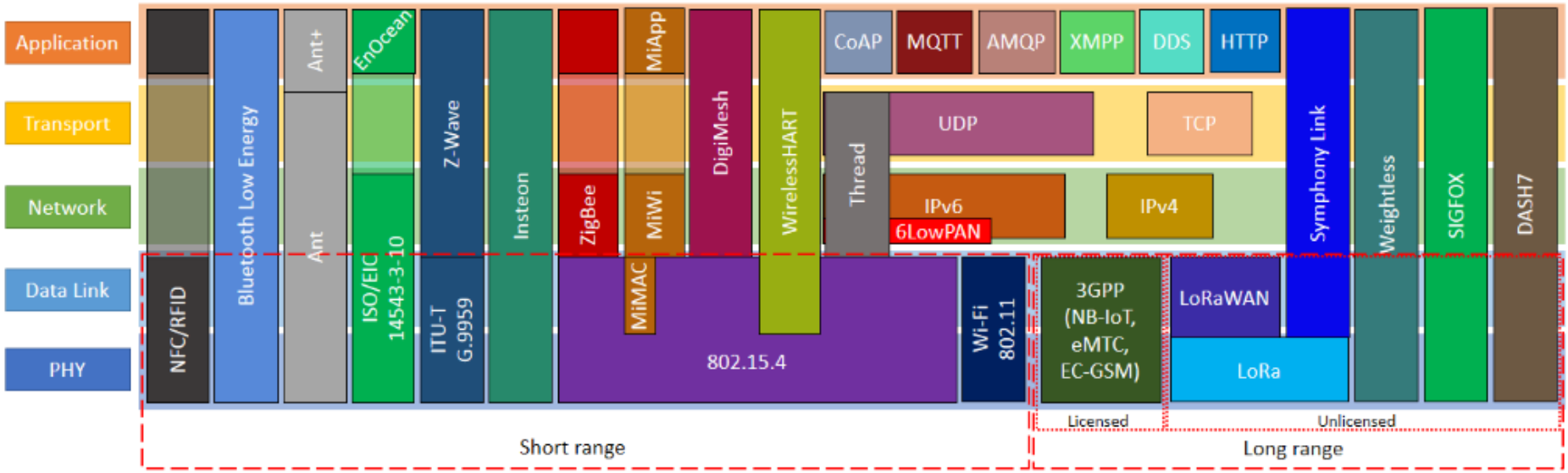
IoT Protocols



IoT Protocols

	IoT Stack		Web Stack
TCP/IP Model	IoT Applications	Device Management	Web Applications
Data Format	Binary, JSON, CBOR		HTML, XML, JSON
Application Layer	CoAP, MQTT, XMPP, AMQP		HTTP, DHCP, DNS, TLS/SSL
Transport Layer	UDP, DTLS		TCP, UDP
Internet Layer	IPv6/IP Routing 6LoWPAN		IPv6, IPv4, IPSec
Network/Link Layer	IEEE 802.15.4 MAC IEEE 802.15.4 PHY / Physical Radio		Ethernet (IEEE 802.3), DSL, ISDN, Wireless LAN (IEEE 802.11), Wi-Fi

IoT Protocols



Short Range Network

- Short range networks use wireless technologies that communicate over ranges of a few centimeters up to hundreds of meters. These include:
  - Close range contactless data transfer technologies,
  - wireless personal area networks (WPANs) as well as,
  - wireless local area networks (WLANs).

Short Range Network

- RFID – Radio Frequency Identification
  - Frequency: Low/High/Ultra-high
  - Range: 1cm - 100m
  - Data Rate: 1-100 kbps
  - Battery: Active (3-5 years)
  - Standardization: Open (no single body)
  - Key Role: Foundation of IoT concept (Ashton, 1999)
  - Applications:
    - Tracking,
    - supply chain,
    - inventory

Short Range Network

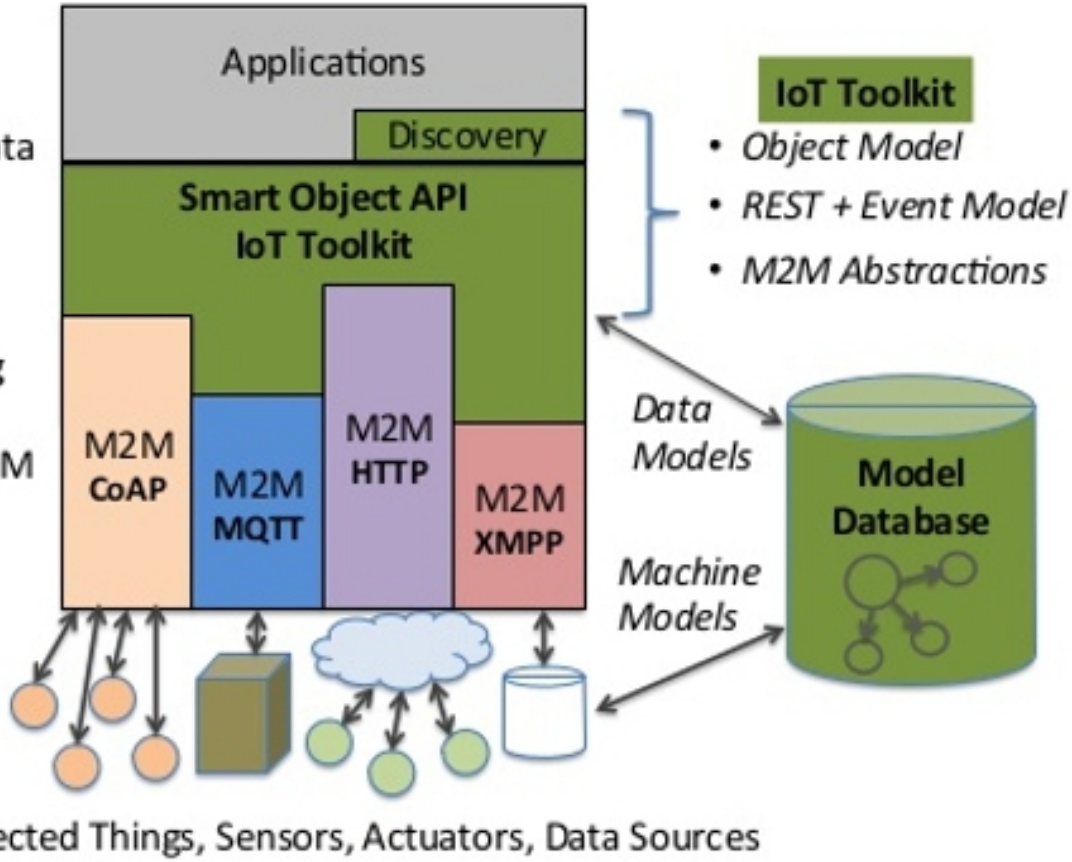
- NFC (Near Field Communication)
  - Frequency: 13.56 MHz
  - Range: ~0.2m
  - Data Rate: 424 kbps
  - Battery: Passive or Active
  - Standardization: ISO/IEC
  - Three Modes: Card emulation, Reader/Writer, P2P
  - Applications:
    - Touch" interactions,
    - pairing,
    - payments

Short Range Network

- BLE (Bluetooth Low Energy)
  - Frequency: 2.4 GHz
  - Range: 10-100m
  - Data Rate: 1 Mbps
  - Battery: Months to years
  - Topology: P2P / Star
  - Standardization: Bluetooth SIG
  - Key Feature: Ultra-low power design

IoT Interoperability What is IoT?

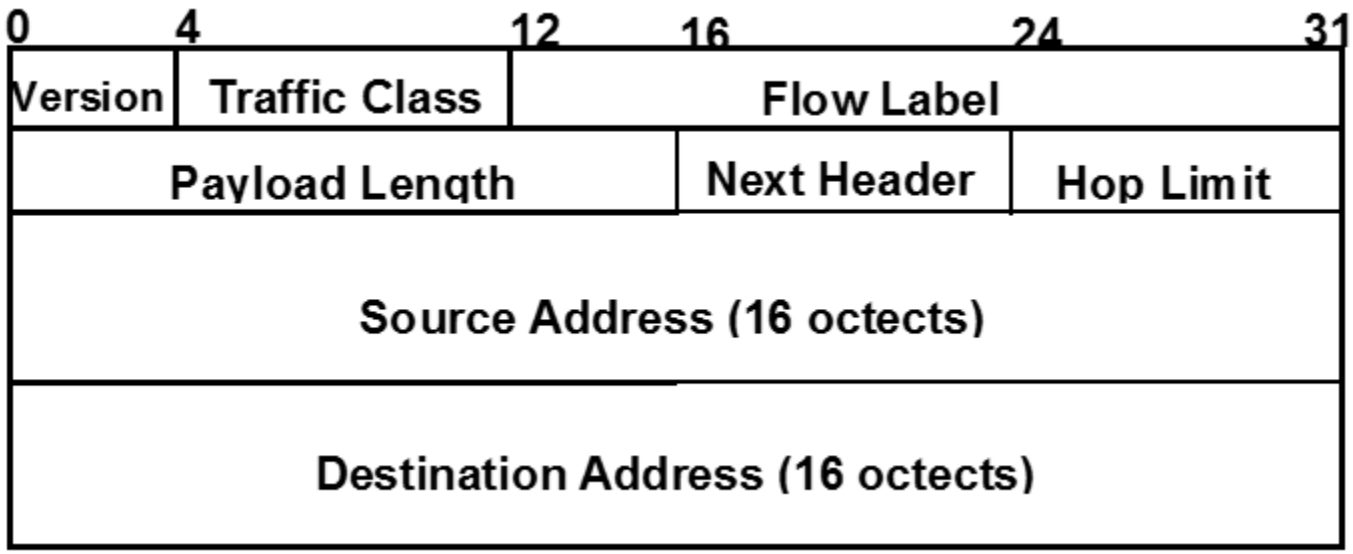
- Easy to deploy new things and applications using data models
- Write once, run anywhere software
- **Any app** to **any thing** via **any M2M**, use-case appropriate M2M
- Network effect enabled



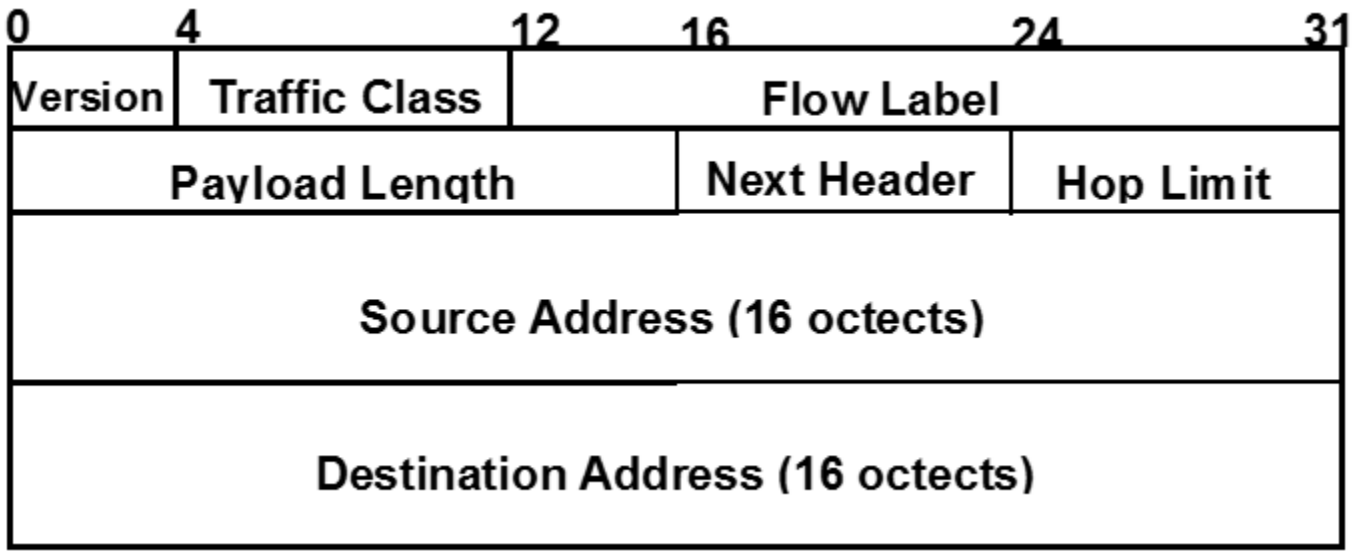
IPv6

- Problems with IPv4
  - Shortage of address space
  - Lack of Quality of Service guarantee
- New features of IPv6
  - Enlarge address space
  - Fixed header format helps speed processing/forwarding
  - Better support for Quality of Service
  - Neighbor discovery and Auto-configuration
  - Hierarchical address architecture (improved address aggregation)
  - new “anycast” address: route to “best” of several replicated servers

IPv6 Header


0
4
12
16
24
31
Version
Traffic Class
Flow Label
Payload Length
Next Header
Hop Limit
Source Address (16 octects)
Destination Address (16 octects)

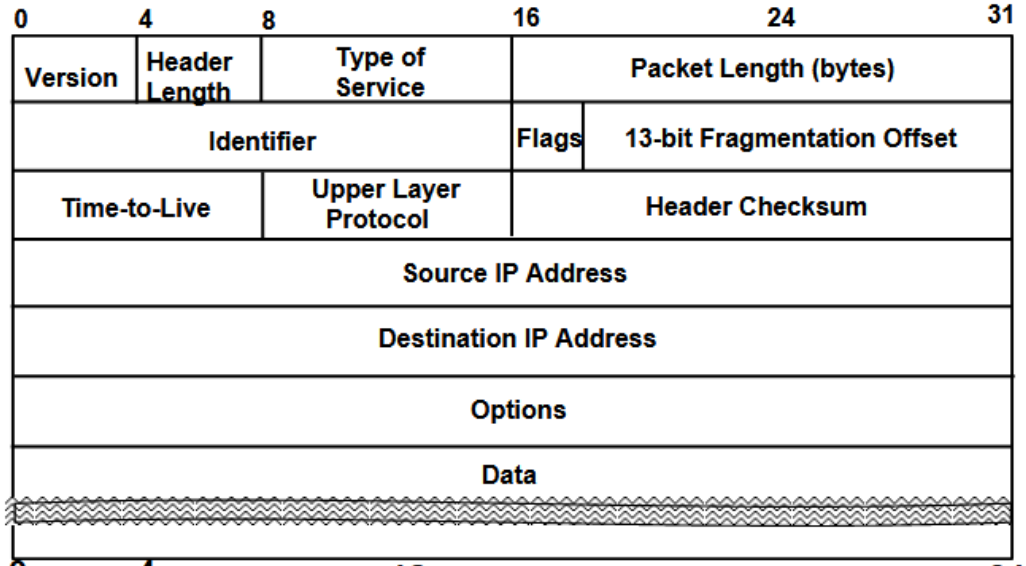
IPv6 Header

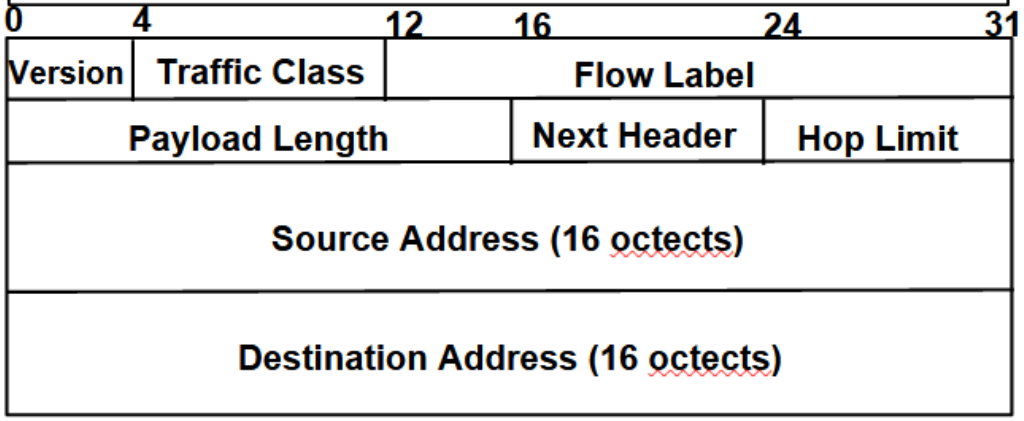

0
4
12
16
24
31
Version
Traffic Class
Flow Label
Payload Length
Next Header
Hop Limit
Source Address (16 octects)
Destination Address (16 octects)

IPv6 Header

- Version: 6
- Traffic class:
  - identify class of service
  - E.g., DiffServ (DS codepoint)
    - The 6 most-significant bits are used for DSCP
- Flow Label:
  - identify datagrams in same “flow”
- Next header:
  - identify upper layer protocol for data

Changes from IPv4





Changes from IPv4

- Expanded Addressing Capabilities
  - from 32 bits to 128 bits (more level and nodes)
  - improve multicast routing (“scope” field)
  - “anycast address”: send a packet to any one of a group of nodes
- Header Format Simplification
  - reduce bandwidth cost
- Extensions
  - more flexibility

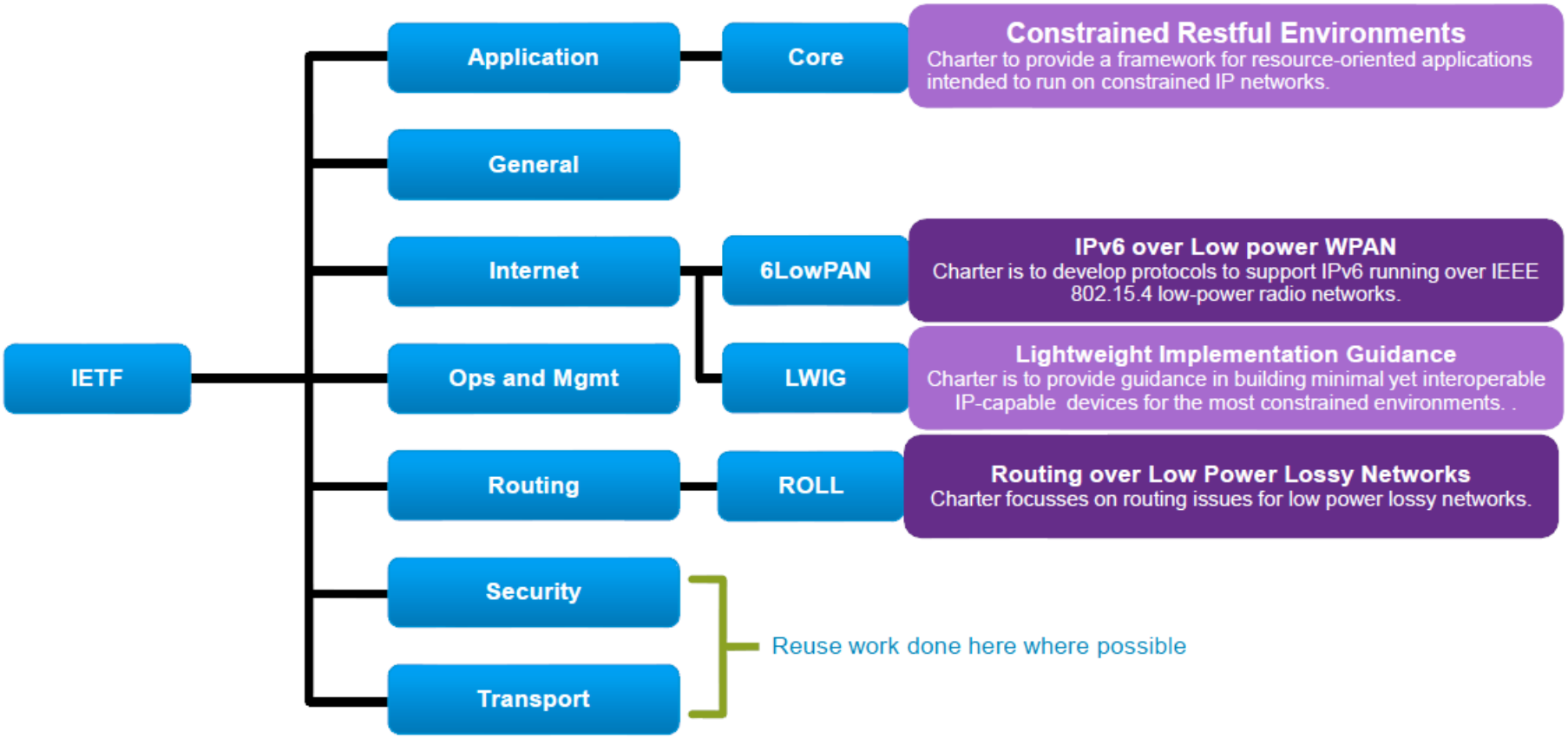
Changes from IPv4

- Checksum
  - removed to reduce processing at routers
- Fragmentation
  - Not allowed at intermediate routers

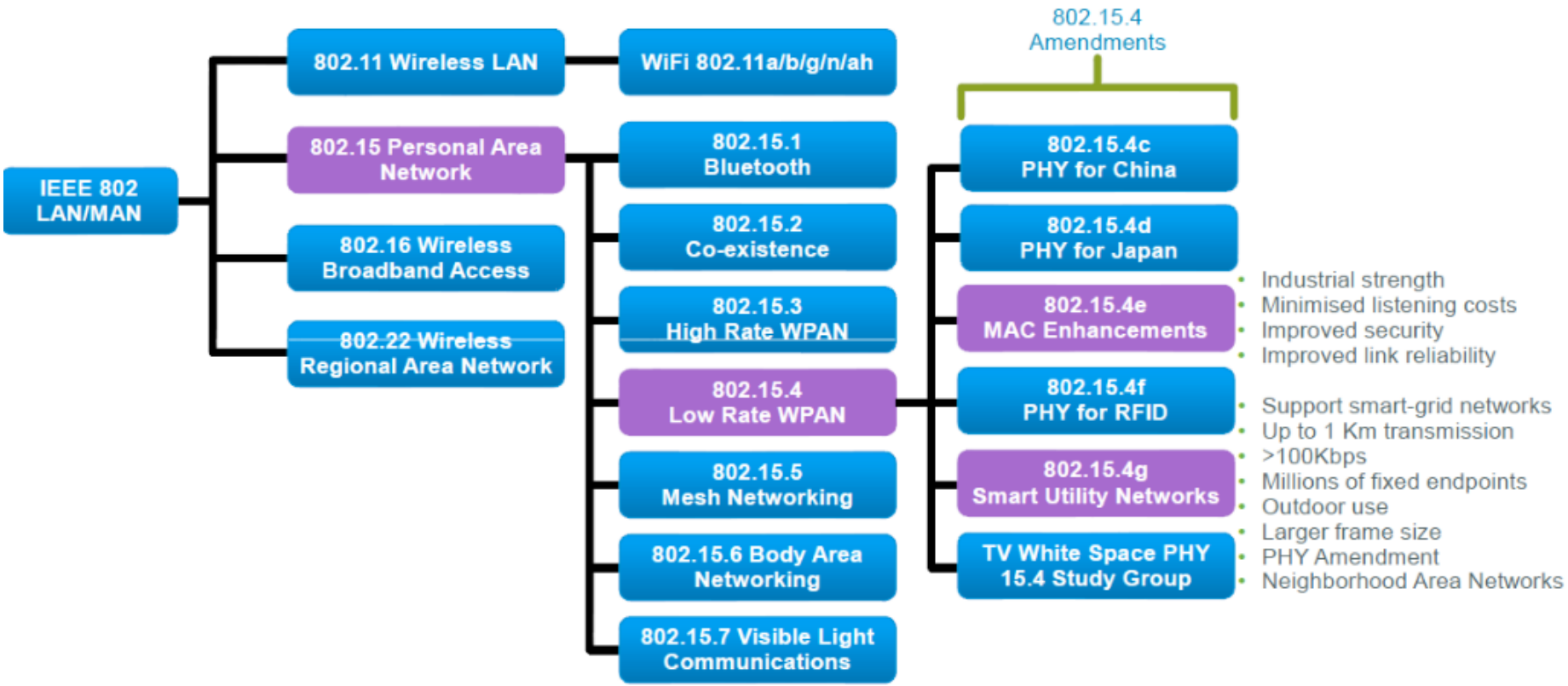
6LoWPAN

- 6LoWPAN is an acronym of IPv6 over Low power Wireless Personal Area Networks.
- It is designed by the 6LoWPAN working group in IETF (Internet Engineering Task Force).
- RFC 4919
- RFC 6282

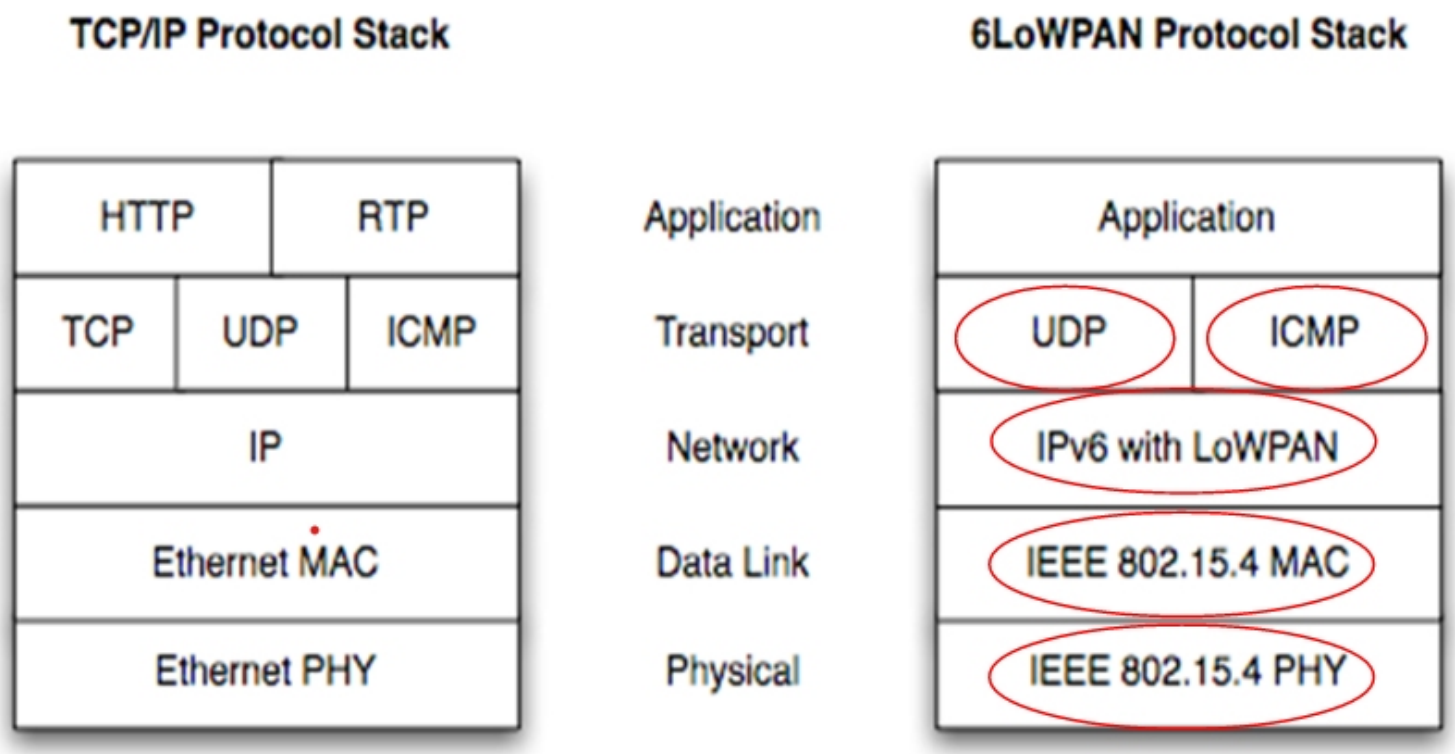
IETF Low Power Lossy Network



IEEE Wireless Standards



IEEE Wireless Standards



6LoWPAN Key Elements

- 6LoWPAN introduces an adaptation layer between the IP stack's link and network layers to enable efficient transmission of IPv6 datagrams over 802.15.4 links
  - Provides header compression to reduce transmission overhead
  - Fragmentation to support the IPv6 minimum MTU requirement
  - Support for layer-two forwarding to deliver and IPv6 datagram over multiple radio hops

RPL

- The IETF Routing Over Low-power and Lossy networks (ROLL) Working Group was formed in 2008
  - to create an IP level routing protocol adapted to the requirements of mesh networking for IoT/M2M
- The first version of RPL (Routing Protocol for Low-power and lossy networks) was finalized in April 2011
- Current standard:

RPL Functionality

- RPL specifies a routing protocol specially adapted for the needs of IPv6 communication over “low-power and lossy networks” (LLNs), supporting
  - peer to peer traffic (point to point) (P2P)
  - point to multipoint (P2MP) communication: from a central server to multiple nodes on the LLN
  - multipoint to point (MP2P) communication
- The base RPL specification is optimized only for MP2P traffic or P2MP, and P2P is optimized only through use of additional mechanisms.

RPL Functionality

- RPL expects an external mechanism to access and transport some control information, referred to as the "RPL Packet Information", in data packets. (ICMP)
- RPL provides a mechanism to disseminate information over the dynamically formed network topology.
  - To reduce the number of messages sent on the network, a trickle algorithm may limit the number of periodic messages that are sent.

RPL Functionality

- In some applications, RPL assembles topologies of routers that own independent prefixes.
- RPL also introduces the capability to bind a subnet together with a common prefix and to route within that subnet.
- RPL may disseminate IPv6 Neighbor Discovery (ND) information such as the Prefix Information Option (PIO) and the Route Information Option (RIO).

CoAP

- The Constrained Application Protocol (CoAP) is defined by IETF CoRE WG for the manipulation of resources on a device that is on the constrained IP networks.

CoAP

- CoAP is
  - A RESTful protocol
  - Both synchronous and asynchronous
  - For constrained devices and networks
  - Specialized for M2M applications
  - Easy to proxy to/from HTTP
- CoAP is not
  - A replacement for HTTP
  - General HTTP compression
  - Separate from the web

Constrained IP Networks

- A constrained IP network has limited packet sizes, may exhibit a high degree of packet loss, and may have a substantial number of devices that may be powered off at any point in time but periodically "wake up" for brief periods of time.
- These networks and the nodes within them are characterized by severe limits on throughput, available power, and particularly on the complexity that can be supported with limited code size and limited RAM per node.
- Low-Power Wireless Personal Area Networks (LoWPANs) are an example of this type of network. Constrained networks can occur as part of home and building automation, energy management, and the Internet of Things.

Devices on Constrained Networks

- The general architecture consists of nodes on the constrained network, called Devices, that are responsible for one or more Resources that may represent sensors, actuators, combinations of values or other information.
- Devices send messages to change and query resources on other Devices.
- Devices can send notifications about changed resource values to Devices that have subscribed to receive notification about changes.
- A Device can also publish or be queried about its resources.

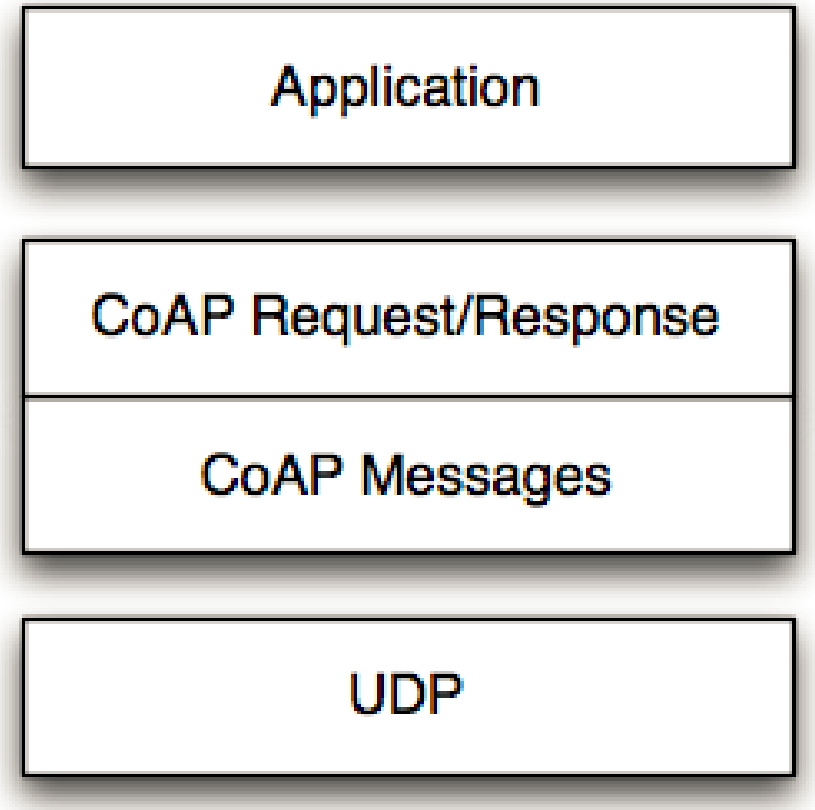
Application Scope of CoAP

- CoAP targets the type of operating environments defined in the ROLL and 6LOWPAN working groups which have additional constraints compared to normal IP networks, but the CoAP protocol will also operate over traditional IP networks.
- This includes applications to monitor simple sensors (e.g. temperature sensors, light switches, and power meters), to control actuators (e.g. light switches, heating controllers, and door locks), and to manage devices.

CoAP vs. HTTP

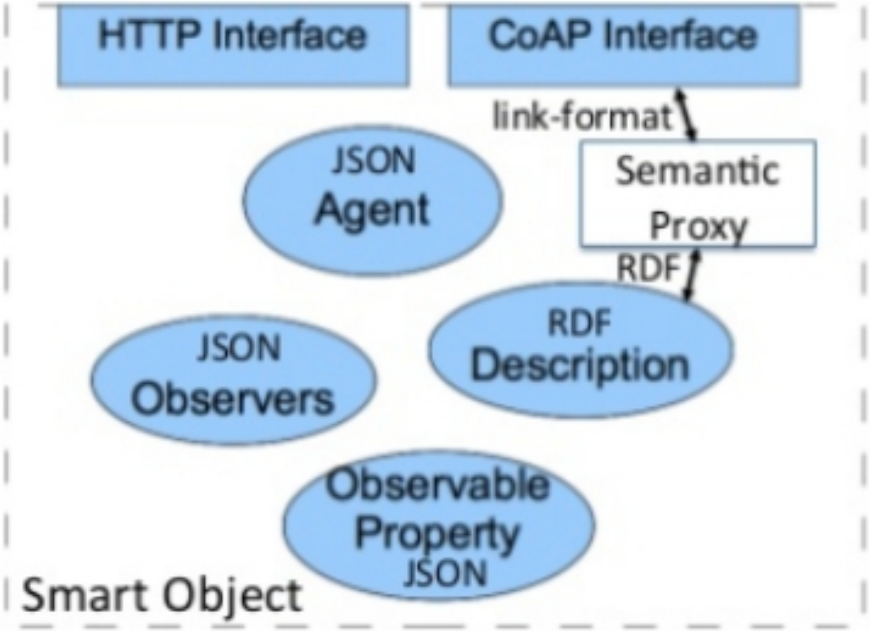
- Like HTTP, the CoAP is a way of structuring REST communications but optimized for M2M applications.
- TCP and HTTP are considered too heavy for 6LowPAN devices such as sensors. CoAP is thus based on UDP and a compressed simplified message exchange.
  - NB: RFC 8323 extends CoAP over TCP/TLS

CoAP RESTful Applications



CoAP Server

CoAP Server Endpoint

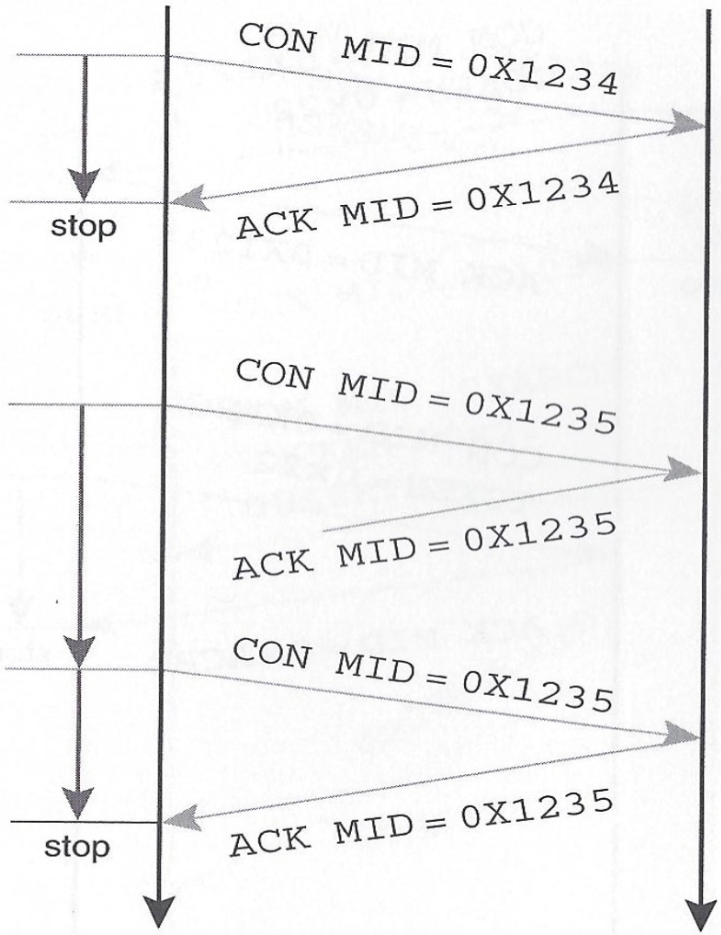


- Compatible Object Models
- Observer pattern
- Standard JSON Resources
- Semantic Proxy exposes stored RDF via core-link-format GET/POST operations

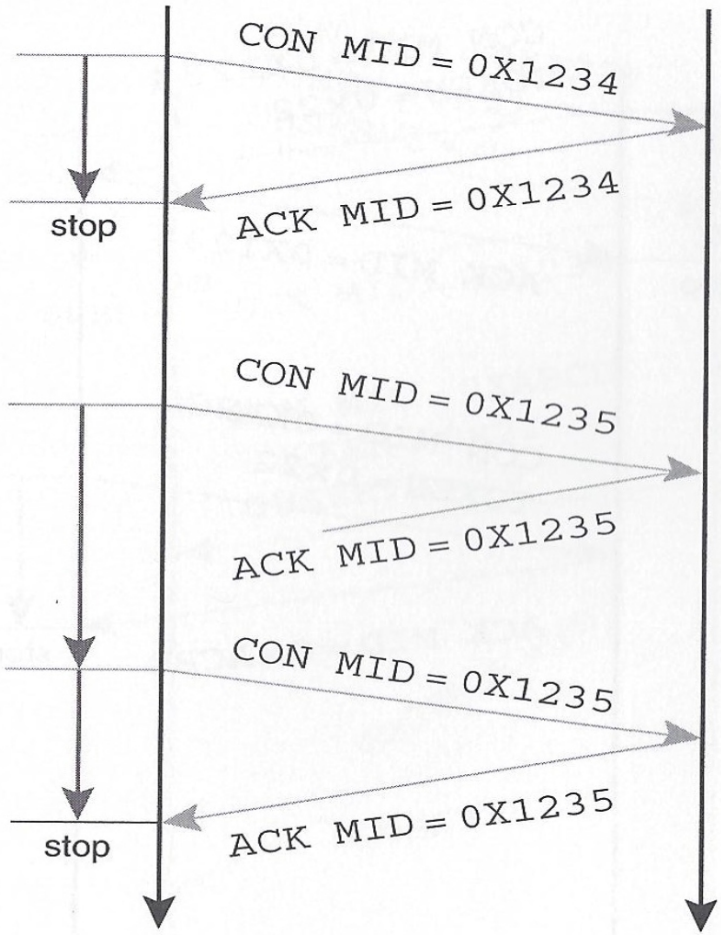
CoAP Methods

- CoAP makes use of GET, PUT, POST, and DELETE methods in a similar manner to HTTP.
- New methods can be added, and do not necessarily have to use requests and responses in pairs.
  - For example: OBSERVE (embedded in GET method)

CoAP Requests



Proxying and Cachings



?

Thanks